



# Transactive memory system, boundary-spanning search and business model innovation: the moderating role of environmental dynamism

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## ABSTRACT

Based on organizational learning theory, the influence of the transactive memory system of the top management team on business model innovation is discussed, as is the impact of different dimensions of boundary-spanning search on different business model innovation attributes when attention distribution is taken into account. Meanwhile, the moderating effects of environmental dynamism are considered. Regression and bootstrap analysis are used to test hypotheses. The results based on empirical research show that the transactive memory system has a positive effect on the dual attributes of business model innovation; acute boundary-spanning search and concentrated boundary-spanning search affect novelty-centered business model innovation and efficiency-centered business model innovation, respectively, and play mediating roles. Meanwhile, environmental dynamism positively moderates the relationship between acute boundary-spanning search and novelty-centered business model innovation, but negatively moderates the relationship between concentrated boundary-spanning search and business efficiency-centered model innovation efficiency.

## 1. Introduction

Business model innovation (BMI) has become a critical battlefield and the goal of enterprise innovation activities and studies calls on enterprises to organize resources and capabilities to achieve business model innovation (Teece, 2010). Top management team (TMT), as important decision makers in the enterprises, have an impact on organizational outcomes and their learning behavior can have a significant impact on business model innovation (BMI) which is one of the most far-reaching decisions. In terms of the subject of action, the impact of top managers' behavior on organizational performance as well as on innovation has been widely demonstrated (Godart et al., 2014; Hambrick and Mason, 1984; Ruiz-Jiménez et al., 2016; Tang et al., 2021). Further, top management behavior plays a critical role in the success of business model innovation (Cillo et al., 2021; Foss and Saebi, 2018). Novelty and efficiency are the two basic attributes of a business model (Zott and Amit, 2007), and novelty-centered and efficiency-centered are two important manifestations of business model innovation in practice (Liu et al., 2022; Yuan et al., 2021). And the discussion of the relationship between business model innovation dual attributes and Top management teams has become a hot issue. The diversity of team members'

roles can bring diverse perspectives to the innovative behavior of a company (Boone and Hendriks, 2009). In particular, the extent to which this diversity triggers differences in internal and external resources can have a differential impact on the novelty and efficiency of business model innovation (Guo et al., 2018). However, the degree of diversity and its threshold is an ongoing topic of debate. In such a context, we use the transactive memory system (TMS) as a perspective to explain the positive effects of diversity.

Organizational learning theory emphasizes organizational performance improvement through learning (Bierly and Chakrabarti, 1996). For business model innovation, learning can be a source (Foss and Saebi, 2017; Sosna et al., 2010). The connectivity property of the business model suggests that the focal firm's simultaneously connects inside the firm and outside the boundaries. At the same time, openness to external innovation knowledge resources helps to improve innovation performance (Katila and Ahuja, 2002; Laursen and Salter, 2006; Leiponen and Helfat, 2010), this means that companies pursuing business model innovation need to stay in close contact with external knowledge. Additionally, the positive impact of TMT's boundary-spanning search on organizations has been proven (Bedford et al., 2022; Li et al., 2013). As one of the key ways of external learning, boundary-spanning search in

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TMT can bring knowledge that is missing internally. However, the choice and intensity of search can have a significant impact on innovation performance (Li et al., 2013). At the same time, there is a correlation between the concentration of TMT boundary-spanning search and the different orientations of business model innovation (Snihur and Zott, 2020). In general, searches that focus on reducing information asymmetry may be beneficial for business model efficiency, while searches that focus on introducing new ideas may be beneficial for creating novel value. Thus, TMT's search for external knowledge may cause different business model innovation results depending on the focus.

One of the goals of learning is to adapt to changes in the environment. The degree of dynamism of the environment is an important context for influencing business decisions, as well as an important external factor influencing the effectiveness of exploratory learning. The innovation of business models of focal enterprises needs to pay close attention to the knowledge of technology, information, and partner dynamism in the external environment outside their boundaries. This knowledge will change as the environment changes. Previous studies have also revealed the different effects of environmental dynamism on the novelty and efficiency of business model innovation (Pati et al., 2018; Waldman et al., 2001; Yuan et al., 2021). Therefore, we expect that the dynamism of the environment will have an impact on the process that boundary-spanning searches affect business model innovation.

There seems to be a lot of discussion about the relationship between internal and external learning. In fact, the debate of alternative versus synergistic roles has been ongoing. Some studies suggest that internal and external learning may be two types of learning that compete with each other for organizational resources (He and Wong, 2004), some studies consider internal learning and external learning as two learning styles that can work together (Gupta et al., 2006; Schroeder et al., 2002) but there is a temporal order of precedence. This study aims to illustrate the different impact paths of internal learning and external learning of TMT on the dual attributes of business model innovation. Under the perspective of business model innovation, i.e., internal learning is considered as the antecedent behavior of external learning. This study used a survey method to collect data from top managers of Chinese "Gazelle Enterprises" in 12 key cities in China through various communication methods. The questionnaire consisted of two parts and was completed by at least three top managers from each firm. The data collected was analyzed using multiple regression and bootstrap analysis to test the measurement and structural models.

The findings from this study make several contributions to the current literature. Firstly, we emphasize the joint effect of TMT's internal knowledge cognition and external knowledge acquisition behavior on business model innovation. The existing literature has limitations on the exploration of TMT factors as the driving factors of business model innovation, and little literature explores the impact of TMT's transactive memory system on business model innovation. Business model innovation focuses more on generic knowledge than technical knowledge (Snihur and Wiklund, 2019; Teece, 2018). And the unique impact of TMT's learning of general knowledge on business model innovation as a decision maker needs to be considered. Concentrating on the black box between TMT's cognitive characteristics background and business model innovation, this study explains how TMT's transactive memory system promote business model innovation through learning behavior. In the context of rapid growing attention to the impact of TMT on business model innovation (Narayan et al., 2021), our research confirms the impact of TMT's transactive memory system on different attributes of business model innovation. Secondly, we integrate the concepts of dual learning (March, 1991) and organizational attention (Ocasio, 2011) into boundary-spanning search and introduce their different effects on the novelty and efficiency of business model innovation. This enriches the literature on organizational learning by considering the relationship between the attention allocation characteristics of search behavior and different attributes of business model innovation. Finally,

we integrate the TMT's perception of existing knowledge, learning behavior, and external environmental factors into a signal framework, propose a new mechanism for transactive memory system affecting business model innovation through boundary-spanning searches and enrich the research on driving factors for business model innovation called by Foss and Saebi (2017).

In terms of managerial impacts, the study suggests that managers can learn several practical implications from this research, such as building and developing transactive memory system for their teams to innovate business models, emphasizing internal learning in top management teams and utilizing specialization, trust, and coordination as responsive forms of transactive memory system, and relying on boundary-spanning search to acquire external knowledge resources for business model innovation. In a context where TMT diversity is pursued but the degree of diversity is difficult to determine, our study provides a new way of thinking for top management teams, reminding them of the importance of transactive memory systems. It also highlights the importance of accurately assessing changes in the external environment and adjusting boundary-spanning search behavior accordingly.

## 2. Theoretical framework and hypotheses

New knowledge that can inspire and inform the business model (Amit and Zott, 2015). This new knowledge can come from multiple sources, such as learning from peers, customers and competitors, or from within the organization (Amit and Zott, 2015; Denoo et al., 2022; McDonald and Eisenhardt, 2020). The connected nature of the business model means that companies can learn internally and externally to gain new knowledge. At the same time, learning behaviors can have different effects on the dual attributes of business model innovation (Yuan et al., 2021). Unfortunately, fewer studies have considered the novel attributes and efficiency attributes of TMT's transactive memory system on business model innovation.

In what follows, we will develop hypotheses about the relationship between the following constructs. Specifically, these include (1) the relationship between transactive memory system and business model innovation, (2) the relationship between boundary-spanning searches and business model innovation and the role of ED, and (3) the ways in which transactive memory system affects business model innovation through boundary-spanning searches.

### 2.1. TMT's transactive memory system and business model innovations

A transactive memory system (TMS) refers to the cognitively interdependent division of labor system, which is used to encode, store, and retrieve information and combine everyone's knowledge with the shared consciousness of "who knows what" (Bachrach et al., 2019; Lewis and Herndon, 2011; Wegner, 1987). As an important form of organizational knowledge repository system, transactive memory system is a latent, reflective structure, involving the integration of knowledge at the group level (Lewis, 2003; Zahra et al., 2020). Team members have their own expertise in a specific field and they tend to rely on each other to obtain, process, and exchange information from different fields (Lewis, 2003; Wegner, 1987). This interdependence enables team members to focus on their professional knowledge in their respective fields and obtain the relevant information required by the task from other members through transactive memory system. Transactive memory system helps to reduce the cognitive load of members, improve learning efficiency, increase the quality and quantity of available information, speed up information search, and reduce the possibility of important information falling through the cracks (Bachrach et al., 2019; Wegner, 1987). Lewis (2003) and Moreland and Myaskovsky (2000) further divided the transactive memory system into three dimensions: specialization, credibility, and coordination. Among them, specialization refers to the differential structure of team members' knowledge, that is, the degree of specialization in different fields of knowledge; credibility refers to the

belief in the reliability of other members' knowledge, that is, the degree of trust members has in each other; and coordination refers to effective and coordinated knowledge processing, that is, the cooperation of members in exchanging knowledge. The transactive memory system of the TMT will have the most significant impact on the whole organization, and this impact is more complex than that of other teams (Heavey and Simsek, 2015). With the enhancement of transactive memory system, TMT would have more knowledge to test and trial the business model. We speculate that transactive memory system has an impact on both attributes of business model innovation.

Business model innovation (BMI) is defined as "designed, novel, and nontrivial changes to the key elements of a firm's business model and/or the architecture linking these elements" (Foss & Saeabi, 2017, 2018), including topics such as complementarity, lock-in, novelty, and efficiency (Zott et al., 2011). In practice, some enterprises have achieved market share through the innovation and disruption of products or services provided. Other enterprises have gained competitive advantage by reducing costs through more efficient operations. Therefore, this study focused on the novelty and efficiency of business model innovation (i.e., BMN and BME), which are the focus of business model innovation research and the most fundamental option for enterprises to create value in a dynamic environment (Yuan et al., 2021). At the same time, they are not mutually exclusive attributes and can exist together (Zott and Amit, 2007, 2008, 2010). Specifically, the novelty of business model innovation is reflected in connecting business model participants in novel ways or designing new transaction mechanisms, while the efficiency-centered business model innovation refers to reducing transaction costs and improving transaction efficiency through new business models (Zott and Amit, 2007, 2008).

Researches have been conducted to highlight the significance of the transactive memory system (TMS) in knowledge-diversified teams (Dai et al., 2016; DeChurch and Mesmer-Magnus, 2010; Lewis and Herndon, 2011; Reagans et al., 2016; Todorova, 2021). The positive effect of the transactive memory system on organizational creativity and innovation performance has been proven (Bachrach et al., 2019). In this study, based on the study of Dai et al. (2016), we consider transactive memory system as an internal learning behavior of TMT. Meanwhile, to compete effectively in a rapidly changing environment, an increasingly important ability of enterprises is to acquire knowledge from the outside, that is, the ability to mine new ideas and technologies from outside their own boundaries (Katila and Ahuja, 2002; Laursen and Salter, 2006; Monteiro and Birkinshaw, 2017; Rosenkopf and Nerkar, 2001). In a dynamic environment, organizations could capture business model innovation by learning to generate new ideas, experiences, and assumptions (Berends et al., 2016; Martins et al., 2015). In this study, we consider boundary-spanning search as an external learning behavior. As a key psychological structure, attention involves the psychological process of paying attention to and interpreting specific clues or signals and investing time and energy to obtain relevant information and knowledge (Li et al., 2013). Obviously, the differences in the allocation of attention for searches may have a variant impact on learning outcomes. Based on the studies of Henttonen and Ritala (2013); Laursen and Salter (2006) this study divides boundary-spanning search into acute boundary-spanning search (AS) and concentrated boundary-spanning search (CS) according to the organization's attention allocation on different knowledge sources and channels. At the same time, according to the function logic of exploratory learning and exploitative learning on business model innovation (Yuan et al., 2021), this study suspects that there may be a correlation between AS and novelty-centered business model innovation (BMN), while there may be a correlation between concentrated boundary-spanning search and the efficiency-centered business model innovation (BME). Organizations rely on variant forms of knowledge repository systems, which enable enterprises to accumulate knowledge based on past experience and affect future performance (Argote et al., 2021; Zahra et al., 2020). With special status and power, the top management team of the enterprise has a strong influence on the

operational activities and strategies of the enterprise. Existing literature also reveals that TMT can search and deploy organizational resources and capabilities to achieve business model innovation (Miroshnychenko et al., 2021; Spieth et al., 2014; Teece, 2010). Therefore, the knowledge that TMT has as a team as a whole is very important for business model innovation.

### 2.1.1. Transactive memory system and the novelty-centered business model innovation

The improvement of the novelty-centered business model innovation (BMN) largely depends on the development and application of new ideas. The transactive memory system expands the cognitive resources available to the team (Heavey and Simsek, 2015) and provides conditions for the novelty of ideas. Specifically, first of all, the diversity of knowledge is the premise of seeking creative combinations (Cohen and Levinthal, 1990). Through the specialization of team members, a transactive memory system can promote the team's understanding of different professional knowledge and skills. Different professional knowledge from different fields can enable members to recognize more valuable business opportunities that independent individuals may ignore (Heavey and Simsek, 2015), which expands the new direction of the team's knowledge vision (Heavey and Simsek, 2017). At the same time, having more complementary sources of knowledge can improve the success rate of innovation (Leiponen and Helfat, 2010). Therefore, transactive memory system helps the team to explore new knowledge and ideas, and then increase the BMN. Second, a strong transactive memory system means that there is higher credibility among members. Due to the effective coordination of team knowledge and trust in each other's professional knowledge, team members have greater personal autonomy in their specific knowledge areas (Dai et al., 2016), which gives them opportunities to explore a wider range of novel business ideas. Third, strong transactive memory system increases the opportunity to discover new ideas. Individual members' more attention to clues means that TMT pays more attention to issues, events, threats, and opportunities collectively (Narayanan et al., 2011). A powerful transactive memory system makes the interaction of TMT members focus on unique and non-overlapping knowledge fields, and the "creative friction" of knowledge in various fields also helps to accelerate the discovery of new knowledge and ideas (Heavey and Simsek, 2017). Together, transactive memory system expands the attention scope of TMT, connects the complementary knowledge fields among members, and makes the subsequent work unique and non-repetitive (Heavey and Simsek, 2017; Zajac et al., 2014), so new knowledge categories and themes pertaining to BMN are more likely to penetrate into the collective consciousness of the team. Previous research has also established that stronger transactive memory system is conducive to the formation of more creative solutions (Ren and Argote, 2011). Therefore, we propose that.

**Hypothesis 1a.** (H1a): TMT's transactive memory system (TMS) has a positive impact on novelty-centered business model innovation (BMN).

### 2.1.2. Transactive memory system and the efficiency-centered business model innovation

The improvement of the efficiency-centered business model innovation (BME) largely depends on the reflection and revision of the existing business model. Transactive memory system focuses on the distinctive expertise of members, that is, the degree of knowledge complementarity among team members (Mohammed et al., 2021). We suspect that this complementarity will promote the innovative utilization and improvement of knowledge resources in relevant fields and promote efficiency innovation. Specifically, first, good recognition and understanding of the existing business model is a necessary condition for business model efficiency innovation (Teece, 2018). The specialization of TMT members means that they can dig deeply into their respective knowledge fields, and their rich experience promotes them to think and reflect on the existing business model, so as to form unique opinions and

share the information with the whole team through transactive memory system. This sharing turns tacit knowledge about the business model into explicit knowledge in the team, which improves the utilization efficiency of organizational knowledge (Kale and Singh, 2007). Second, the credibility and coordination among TMT members improve the efficiency of team knowledge flow and complex information processing (Lewis and Herndon, 2011). It reduces the communication costs between top management members. More efficient communication also means that the organization can more accurately locate and quickly solve the problems faced by the current business model. Third, transactive memory system can integrate members and tasks more effectively (Reagans et al., 2016). By consummating the division of tasks and reasonably matching personnel and tasks, organizations can maximize the use of limited resources and achieve effective internal coordination, which in turn reduces the transaction cost of enterprises. Together, transactive memory system enhances the BME by facilitating the reflection and improvement of the existing business model. Therefore, we propose that.

**Hypothesis 1b.** (H1b): TMT's transactive memory system (TMS) has a positive impact on efficiency-centered business model innovation (BME).

## 2.2. Transactive memory system and boundary-spanning searches

Search is defined as the controlled and proactive process of attending to, examining, and evaluating new knowledge and information (Li et al., 2013). It is an important form of organizational learning. Knowledge-based innovation literature also suggests that search is a necessary process (Cohen and Levinthal, 1990; Grant, 1996; Trantopoulos et al., 2017). Studies have confirmed the applicability of boundary-spanning search at the individual level, team level, and organization and cross-organization level. For example, members of different work teams (Ancona and Caldwell, 1992; Brennecke et al., 2021; Dahlander et al., 2016; Marrone et al., 2022; Rosenkopf and Almeida, 2003), different work teams within an organization (Hansen, 1999; Marrone, 2010; Marrone et al., 2022), and different organizations (Monteiro and Birkinshaw, 2017; Posen et al., 2018; Rosenkopf and Almeida, 2003; Rosenkopf and Nerkar, 2001; Wang and Gao, 2021). In this study, boundary-spanning search is defined as the process by which TMT connects with the environment outside the organizational boundary and acquires knowledge in order to achieve organizational goals. It has been considered that breadth and depth are the two main characteristics of search. Katila and Ahuja (2002) defined search breadth and search depth as the degree of exploring new knowledge and using existing knowledge, respectively. Laursen and Salter (2006) divided external search into search breadth and search depth according to the concentration of external resources used. This division has been widely used (Leiponen and Helfat, 2010). Based on the above discussion, we divide boundary-spanning searches into two types: acute boundary-spanning search (AS) and concentrated boundary-spanning search (CS). Specifically, AS is defined as a search process in which organizations concentrate on a wider range of knowledge sources in order to comprehensively obtain external knowledge. CS is defined as a search process in which knowledge sources from a small number of channels are searched to deeply obtain external knowledge. AS means timely and extensive acquisition of market, technology, policy, and other knowledge in multiple fields beyond the boundaries of the organization. CS means focusing on and deeply capturing the market, technology, policy, and other knowledge in specific fields beyond the boundaries of the organization. Most of the core decisions and actions of the TMT, such as market demand and technical demand, are inseparable from externally driven requirements (Heavey and Simsek, 2015). The connection attribute of the business model requires TMT to pay close attention to external demand and competitors in order to promote feasible innovation. At the same time, at the team level, team performance behavior is

the result of transactive memory system rather than the antecedent (Ren and Argote, 2011). Studies have suggested that transactive memory system can promote the information acquisition behavior of organizational learning (Bachrach et al., 2019; Lewis and Herndon, 2011; Wegner, 1987). Therefore, it seems possible that the team knowledge base as a cognitive feature will affect the learning behavior of TMT. That is, the transactive memory system may have an impact on boundary-spanning searches.

### 2.2.1. Transactive memory system and acute boundary-spanning search

Transactive memory system can expand the team's domain of knowledge exploration (Heavey and Simsek, 2017), facilitating acute boundary-spanning search (AS). Specifically, first, transactive memory system provides the possibility to expand the source of knowledge by encouraging members' diverse exploration. A well-developed transactive memory system means that team members agree that their diversity can bring complementary and valuable information to the team (Ren and Argote, 2011; To et al., 2021; van Knippenberg et al., 2013). This awareness of team diversity legitimizes individual sharing rather than suppresses individual unique views (van Knippenberg et al., 2013). In this situation, members have more freedom to express their unique views and are more willing to introduce diversified information to the team through AS. Secondly, powerful transactive memory system enables TMT to pay attention to the channels of multiple knowledge sources at the same time. Through the professional labor division, top managers with strong transactive memory system have a more inclusive vision when scanning the environment (Heavey and Simsek, 2017). The division of labor brought by well-developed specialization enables members to become experts in their respective fields and to be more alert to changes in knowledge in that field, which enhances the timeliness of external knowledge acquisition. The in-depth knowledge and understanding of their own knowledge fields can also enable team members to pay close attention to the latest external changes in the information, technology, and other knowledge so that the team can keep the overall knowledge base up to date. Finally, transactive memory system allows for a smoother exchange of knowledge within TMT. By understanding and trusting each other's knowledge, the team as a whole is able to flow knowledge smoothly and quickly, which reduces the search burden on members and brings positive feedback to the broader search. With high credibility and coordination, members have more energy to proactively expand their search scope and pay close attention to the external heterogeneous knowledge that may contribute to the expansion of the organization's overall knowledge. Therefore, we propose that.

**Hypothesis 2a.** (H2a): TMT's transactive memory system (TMS) has a positive impact on acute boundary-spanning search (AS).

### 2.2.2. Transactive memory system and concentrated boundary-spanning search

It seems possible that transactive memory system could promote concentrated boundary-spanning search (CS). Specifically, first, transactive memory system can further increase the depth of the members' search field. Transactive memory system relieves members of the burden of learning and reviewing information in multiple areas, enabling them to acquire more in-depth and specialized knowledge (Moreland and Myaskovsky, 2000; Wegner, 1987). Specialization means that members have in-depth knowledge of their respective fields, and through the accumulation of knowledge, knowledge inertia is formed, which drives members to continue to explore further in their areas of knowledge (Lewis, 2003). At the same time, exploring in familiar fields is inherently more efficient (Laursen and Salter, 2006). This efficient learning feedback further enhances the concentration of members' search. Therefore, members can devote more attention to familiar areas. Second, credibility and coordination lead to a common understanding of the team's search objectives through adequate communication and trust in each

other's search actions. Transactive memory system facilitates team members to search and locate exactly what they need to know and to solve problems efficiently by matching tasks with members who have the relevant expertise (Moreland and Myaskovsky, 2000), so as to make more effective use of the contributions of individual members (Reagans et al., 2016). The recognition of individual contributions will enhance their search self-efficacy, which boosts members' confidence in exploring their original areas of knowledge. The powerful transactive memory system promotes the team's overall understanding of the knowledge in task-related fields, and the coordination makes the knowledge exchange between team members smoother, with the resulting more efficient communication reducing the effort that can be expended on subsequent knowledge integration. Therefore, coordination allows members to focus on their tasks for more in-depth boundary-spanning searches. Individuals may lack the understanding of knowledge in other fields due to their excessive attention to their own fields, and the team cooperation brought by transactive memory system makes members concentrate on their respective fields by getting the cognitive support of the team as a whole, which is helpful for the team to further explore the knowledge in specific fields. Based on the above analysis, we propose that.

**Hypothesis 2b.** (H2b): TMT's transactive memory system (TMS) has a positive impact on concentrated boundary-spanning search (CS).

### 2.3. Boundary-spanning searches and business model innovation

#### 2.3.1. Acute boundary-spanning search and novelty-centered business model innovation

Novelty of business model innovation is reflected in connecting business model participants in new ways or designing new transaction mechanisms (Zott and Amit, 2007, 2008). This requires a sufficient number of alternative options to solve the problem. Acute boundary-spanning search (AS) means paying attention to many fields, including unknown fields, and paying timely and extensive attention to the knowledge of technological innovation, product demand, market changes, and business model innovation practices of other industries. It is expected that AS can positively affect BMN. Specifically, first of all, AS enables novel ideas to be realized ahead of competitors through a high degree of timeliness in knowledge acquisition. TMT's acute search enables organizations to understand the changes in external demand and supply timely. When changes in the competitive environment present opportunities for innovation, AS enables enterprises to notice them ahead of competitors. Time lead allows companies to apply new ideas and knowledge to product development and strategic design ahead of competitors (Laursen and Salter, 2006; Lopez-Vega et al., 2016; Snihur and Wiklund, 2019). This allows organizations to provide new products or services ahead of their competitors. Second, AS provides a wider range of external heterogeneous knowledge to facilitate the novelty of business model innovation. Innovation is associated with a wide range of knowledge sources (Leiponen and Helfat, 2010). And the exploration of unrefined areas often leads to novelty (Ahuja and Morris Lampert, 2001; March, 1991; Rosenkopf and Nerkar, 2001). Therefore, the new experience and knowledge gained from exploratory learning can help to break through the barriers of ambiguity and promote BMN. This is also because access to a wider range of knowledge enriches the organization's knowledge repository, generating new ideas through creative friction with existing knowledge. The updated knowledge repository allows enterprises to redefine, select or add novel content to business models (Mcmillan, 2015; Nerkar, 2003; Yuan et al., 2021). At the same time, AS means that TMT members track demand changes in a timely manner, pay attention to policy information and partner dynamism, and remain alert to emerging opportunities in the market. This search behavior facilitates the formation of ideas by promoting the breadth and timeliness of heterogeneous knowledge sources. Research by Snihur and Wiklund (2019) also found a strong association between broad external

searches and the novelty of business model innovation. Therefore, we propose that.

**Hypothesis 3a.** (H3a): Acute boundary-spanning search (AS) has a positive impact on the novelty-centered business model innovation (BMN).

#### 2.3.2. Concentrated boundary-spanning search and the efficiency-centered business model innovation

The efficiency-centered business model innovation (BME) aims to reduce transaction costs and improve transaction efficiency through new business models (Zott and Amit, 2007, 2008). Concentrated boundary-spanning search (CS) focuses on a few specific knowledge fields, including familiar ones, and pays close attention to the knowledge of technological innovation, product demand, market changes, and policies. It is expected that this focus is conducive to the BME. First, CS improves the learning efficiency of external knowledge acquisition. An in-depth search in a certain field can make the search more reliable (Levinthal & March 1981). Because concentrated search means that repeated efforts are invested in a particular domain, the knowledge acquired has a high degree of similarity or consistency. The higher the consistency of knowledge sources, the more conducive it is for enterprises to strengthen experiential learning, improve learning depth, promote incremental innovation, optimize transaction models, and ultimately improve transaction efficiency (Yu et al., 2020). Second, CS adds depth to the understanding of current business models. BME requires good familiarity with and understanding of current business model practices (Teece, 2018). CS facilitates organizational reflection on the content, structure, and governance of current business models, which can make knowledge flow between stakeholders smoother and reduce information asymmetry, as well as help enterprises to take full account of stakeholder aspirations and establish good relations with business partners. Third, reliable exploration of specific fields also helps to correctly judge the feasibility of an opportunity. Focusing on a specific domain of knowledge means exploring the empirical logic behind it while exploring possible opportunities in-depth and developing a more accurate perception of the market. For example, clear and thorough insights into the market help to upgrade existing products or launch new ones to meet market demand and expand market share (Martin and Mitchell, 1998). At the same time, borrowing and learning from the target industry can free the organization's attention to innovate in more valuable areas (McDonald and Eisenhardt, 2020), reducing the risk of business model innovation failure. Together, a deeper search in familiar domains facilitates following existing learning experiences, promotes exploitative learning, and fosters efficiency-centered business model innovation (Yu et al., 2020). Therefore, we propose that.

**Hypothesis 3b.** (H3b): Concentrated boundary-spanning search (CS) has a positive impact on the efficiency-centered business model innovation (BME).

### 2.4. The mediating effects of boundary-spanning searches

In order to compete effectively in a rapidly changing environment, the ability to acquire new knowledge and technologies from beyond the boundaries of the enterprise is increasingly important (Katila and Ahuja, 2002; Laursen and Salter, 2006; Rosenkopf and Nerkar, 2001). Active exploration of knowledge in the external environment is a vital means of heterogeneous knowledge acquisition. By exploring heterogeneous knowledge beyond organizational boundaries, firms can combine new knowledge with existing knowledge to develop new knowledge combinations that in turn promote business model innovation. However, boundary-spanning search is inseparable from the influence of the existing knowledge repository. The knowledge-based innovation theory shows that different types of innovation require different knowledge sources (Cohen and Levinthal, 1990; Trantopoulos et al., 2017), which means that the novelty and efficiency of business model innovation have

different knowledge sources demand. Therefore, it seems that the transactive memory system of TMT will affect different attributes of business model innovation through different forms of boundary-spanning searches.

TMT's cognition of an existing knowledge repository can influence business model innovation by influencing the process of acquiring heterogeneous knowledge through exploratory learning behavior across organizational boundaries. Having different areas of expertise encourages a broader search for solutions, facilitates the exchange of ideas, stimulates the exploration of task-relevant information (van Knippenberg et al., 2004), and increases opportunities for solving particularly challenging problems (Vestal and Mesmer-Magnus, 2020). A strong transactive memory system expands the fields of expertise of the TMT through specialization, credibility, and coordination, enabling members to explore knowledge beyond the boundaries of the organization, thereby increasing the possibility of business model innovation. Existing research has highlighted transactive memory system as a key driver of team learning (Lazar et al., 2021; Ren and Argote, 2011). Meanwhile, the potential beneficial effects of transactive memory system are inseparable from the team's external knowledge sources and needs (Heavey and Simsek, 2015). This suggests that transactive memory system can further influence business model innovation by influencing the inflow of external knowledge. In addition, the study by Snihur and Zott (2020) also suggests that the mindset of top managers affects the scope of the search and therefore business model innovation. Transactive memory system helps reduce the duplication of members' cognitive actions and then expands their attention to a wider field, obtains external heterogeneous knowledge in a timely and extensive manner, generates novel ideas ahead of their opponents, and promotes the BMN. At the same time, transactive memory system can improve the efficiency of knowledge flow among TMT members, facilitating members to focus on their respective knowledge fields, gaining deeper access to external heterogeneous knowledge, and reflecting on existing business models to improve transaction efficiency. Based on the above discussions, we expect that AS can mediate the relationship between transactive memory system and BMN, while CS can mediate the relationship between transactive memory system and BME. Therefore, we propose that.

**Hypothesis 4a.** (H4a): Acute boundary-spanning search (AS) mediates the relationship between transactive memory system (TMS) and novelty-centered business model innovation (BMN).

**Hypothesis 4b.** (H4b): Concentrated boundary-spanning (CS) mediates the relationship between transactive memory system (TMS) and the efficiency-centered business model innovation (BME).

## 2.5. The moderating effects of environmental dynamism

Boundary-spanning search is an important means to connect internal and external knowledge of an enterprise, and changes in external knowledge caused by environmental dynamism will affect the performance of boundary-spanning search. Therefore, it seems that the strength of the relationships between boundary-spanning searches and business model innovation would be affected by environmental conditions. Environmental dynamism (ED) reflects the rate of change and instability of the external environment (Dess and Beard, 1984), including the dynamism of markets, technologies, etc. (Dess and Beard, 1984; Zhang and Zhu, 2021). Low environmental dynamism is characterized by small changes and most market behaviors are predictable (Dubey et al., 2021; Sirmon et al., 2007); conversely, high environmental dynamism is characterized by high volatility, with rapid, persistent change (Schilke, 2014). Changes in information, resources, and knowledge in the environment affect the operation and innovation of business models. For higher environmental dynamism, it is expected that AS is more likely to bring novelty to business model innovation for the following reasons. First, in a highly dynamic environment, the rapid

changes in market demand provide more space for acute boundary-spanning search (AS) to discover opportunities. At the same time, in a dynamic environment, opportunities are fleeting, threats are imminent, and delays are extremely costly (Girod and Whittington, 2017). Therefore, AS is more likely to enable enterprises to grasp the changes in the relationship between business model elements in time, so as to apply new ideas to them and achieve novel innovations ahead of their competitors. Second, AS targets broad heterogeneous knowledge in the external environment. High environmental dynamism indicates high environmental variability (Duncan, 1972), which also means drastic changes in external knowledge supply, so there is necessarily a great deal of novel and heterogeneous knowledge in it. Furthermore, environmental changes often bring about previously unavailable or unforeseen opportunities, which can be linked to more creative business models (McDonald et al., 2021). BMN's need for extensive heterogeneous knowledge increases the value of AS. That is, in the case of high environmental dynamism, the more timely and broader knowledge sources provided by AS would have greater value. That provides a more sufficient basis for the organization's business model innovation decisions and increases the possibility of transforming novel ideas into successful practice. In the lower dynamic environment, the external technology, market, and policy environments are relatively stable, and the energy expended in AS cannot effectively bring novel ideas to enterprises to promote BMN. Therefore, AS in a highly dynamic environment is more important for the BMN. Based on the above discussion, we propose that.

**Hypothesis 5a.** (H5a): High environmental dynamism (ED) enhances the relationship between acute boundary-spanning search (AS) and novelty-centered business model innovation (BMN).

Dynamic environments bring uncertainty and lead to organizational environments characterized by stress, anxiety, and risk (Waldman et al., 2001). At the same time, enterprises' behavior lags behind environmental changes, so a highly dynamic environment may have a negative impact on corporate operations (Bao et al., 2020). Frequent changes in products, customer preferences, and technology can quickly erode the effectiveness and efficiency of operational routines (Wilhelm et al., 2015). Therefore, it is expected that the increased dynamism of the environment may negatively affect the relationship between concentrated boundary-spanning search (CS) and the efficiency-centered business model innovation (BME). Specifically, first, the search performance of CS is negatively affected by environmental dynamism. In a highly changing environment, existing products and services are prone to becoming outdated (Bao et al., 2020). Large changes in knowledge supply and demand can make it difficult to acquire valuable knowledge through specific channels according to established practices. This increases the search cost of acquiring valuable heterogeneous knowledge. Second, dynamic environments force firms to perform costly and irreversible reallocations of existing resources (Yuan et al., 2021). And frequent reconfigurations may increase transaction costs and undermine the efficiency of business model design (Amit and Zott, 2015; Song et al., 2006). At the same time, environmental dynamism may cause dominant practices in the industry to fail or cause confusion so as to increase costs (Yuan et al., 2021). This also means that the knowledge gained by spanning organizational boundaries and spending a lot of resources can hardly match the cost to the enterprise. Third, high environmental dynamism negatively impacts the value of existing knowledge channels. The knowledge channel established by CS has high stability. With relatively little change in the environment, this stability can provide a continuous and effective source of knowledge to the organization and help improve the BME. However, as the environment becomes more dynamic, this stability comes with enormous inertia. High environmental dynamism rapidly reduces the value contribution of established operating practices (Winter, 2003). This means that the value of existing knowledge channels would be reduced, and the effect of CS on BME would be weakened. Because of the lag between corporate decisions and

the environment, existing partnerships that have taken a lot of effort to establish may not be able to adapt to the dramatic changes in the environment when the environment changes significantly. Continuing to focus on existing sources of knowledge makes it difficult for organizations to advance business model changes to gain efficiency advantages over competitors. Therefore, we propose that.

**Hypothesis 5b.** (H5b): High Environmental dynamism (ED) weakens the relationship between concentrated boundary-spanning search (CS) and the efficiency-centered business model innovation (BME).

Incorporating the above hypotheses, moderated mediation hypotheses could be proposed that environmental dynamism moderates the indirect effects of transactive memory system on business model innovation via boundary-spanning searches. Specifically, the transactive memory system of TMT under high environmental dynamism is more likely to achieve BMN through AS, while the transactive memory system under low environmental dynamism is more likely to achieve BME through CS.

**Hypothesis 6a.** (H6a): Environmental dynamism (ED) moderates the indirect relationship of transactive memory system (TMS) affecting BMN via AS; that is, the indirect relationship is more positive when environmental dynamism is higher.

**Hypothesis 6b.** (H6b): Environmental dynamism (ED) moderates the indirect relationship of transactive memory system (TMS) affecting BME via CS; that is, the indirect relationship is more positive when environmental dynamism is lower.

### 3. Methods

The previous section shows the variables of the study and the hypothesis of the relationship between the variables. Because the variables in the above hypotheses are difficult to observe directly, we use response-based indicators to indicate the above variables. We tested the measurement model according to Hair et al. (2021) and tested the structural model through multiple regression and bootstrap analysis.

#### 3.1. Samples and data collection

We collected the perceptions of the top managers of the enterprises on the subject items by means of a questionnaire. Specifically, we collect data on Chinese “Gazelle Enterprises”, which are characterized by high growth (Acs and Mueller, 2008; Lindic et al., 2012), strong demand for knowledge (Croce et al., 2021), and innovation in technology or business models to achieve growth. This provides a good research laboratory for our business model innovation research. Specifically, we collected samples from enterprises in 12 key cities, including Beijing, Shanghai, Guangzhou, Xi'an, Hangzhou, Jinan, and Chengdu, etc. These biggest cities cover the regions with the most active economic growth in China. These cities in the top 30 economies in China take a total GDP share of over 40%. The total number of gazelle enterprises in these cities is close to 40% nationwide, representing various economic regions, and therefore can provide a representative sample for this study. The enterprise directory was derived from the list of gazelle enterprises in 2020 issued by the local government. 240 enterprises in the fields, healthy, artificial intelligence and manufacturing, enterprise services, and retail e-commerce and new brands were randomly selected. In these fields, business model innovation is a more important support for growth. By using telephone, email, instant messaging software, and social media, they were invited to participate in the survey, and a total of 176 companies' top management teams agreed to participate in it. This represents 1% of the total number of gazelle enterprises in these regions.

Through a series of procedures, we tried to ensure the validity and reliability of the questionnaire. First, we designed a questionnaire in English and translated it into Chinese through double and reverse translation procedures. The management school of a top university in

China, to which the researcher belongs, has a wide range of MBA and EMBA students, and a large number of them serve as top managers in enterprises. We invite some participants to read our questionnaire and provide us with feedback. Second, this research conducted a field trial survey. Questionnaires were distributed on-site at the Changyang Chuanggu National Innovation and Entrepreneurship Campus in Shanghai. With the assistance of the management department, five enterprises with high growth characteristics were randomly selected, and the top managers of the enterprises were asked to fill in the questionnaire on-site, and their feedback on the content of the questionnaire, including terminology, language item wording, and content, was collected. Then, according to the feedback, the content of the questionnaire was supplemented and revised after the discussion of three professional researchers. After this, we proceed to further large-scale distributions.

The content of the questionnaire consisted of two parts, A and B. We urged the surveyed firms to have as many TMT members as possible participate in our survey and to ensure that at least three top managers in each firm are involved in the survey. For questionnaires, part A focuses on the transactive memory system at the team level, with at least two top managers from each firm completing their respective responses; Part B deals with the basic information of firms, the boundary-spanning searches, business model innovation, and the dynamism of the environment, and was completed by another top manager or team leader who was not involved in part A. The two parts of the questionnaire were numbered by enterprises for matching and together form the complete questionnaire.

The questionnaire collection process spanned 45 days, from July 2021 to August 2021. We alerted companies that did not respond on the 15th and 30th days of the survey. At the end of the survey, a total of 137 responses were received, with a team-level response rate of 77.84%. Questionnaires were excluded where the respondent's role did not match, where there were missing content and obvious errors in common sense content, of which 97 were for part A and 30 for part B. At the same time, data from the sample with less than 2 valid completers in part A were excluded, and a total of valid survey data from 87 teams was obtained, including 87 valid questionnaires for part A and 276 corresponding valid questionnaires for part B. The valid response rate for the overall sample was about 52.3%. An average of 4.17 managers responded per copy. Referring to related studies (Bachrach et al., 2017; Bachrach and Mullins, 2019), the number of samples provided the basis for our further analysis.

We examined potential no-response or late-response bias according to Armstrong and Overton (1977). Among unanswered and answered responses, there were no significant differences in firm industry ( $p = 0.789$ ), firm age ( $p = 0.243$ ), and firm ownership type ( $p = 0.230$ ). Among the answers, there were no statistically significant differences between early-answers and late-answers in industry ( $p = 0.453$ ), firm age ( $p = 0.480$ ), ownership type ( $p = 0.387$ ), firm size ( $p = 0.762$ ), and number of top managers ( $p = 0.455$ ). This suggests that non-response bias and post-response bias will not have a significant impact on the study.

In our valid sample, foreign-owned enterprises accounted for 8.05%, private-owned enterprises accounted for 42.53%, mixed-ownership enterprises accounted for 28.74%, and other ownership enterprises accounted for 20.69%. The industry distribution was healthy (20.7%), artificial intelligence and manufacturing (24.1%), enterprise services (28.7%), and retail e-commerce and new brands (26.4%). In valid responses, the top management teams had a minimum of 3 members and a maximum of 11 members. For establishment years of enterprises, 87.4% were established within 10 years.

#### 3.2. Measures

To ensure reliability and validity, most of the scales in this study used widely validated scales. At the same time, to ensure the applicability of

the scale, we invited professional researchers and top managers to revise the scales according to the research background. All scales were measured utilizing a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The scales and items are as follows.

**Independent variable: Transactive memory system (TMS).** We used the scale developed by Lewis (2003) and adjusted by Heavey and Simsek (2017). The original scale contains 15 items in 3 dimensions. After adjusting for the research needs, this study used 9 items to measure the transactive memory system. The items are: “Each member has expertise in some aspect of our work”, “Achieving our goals requires the expertise of several team members”, “I know which team members have expertise in a particular area”, “I am open to suggestions from other team members”, “I trust the knowledge of other team members to be reliable”, “I have confidence in relying on information brought by other team members”, “Our team works well together”, “Our team has little misunderstanding about what to do”, and “We tend to get things done smoothly and efficiently”. Since the transactive memory system is at the team level, the data collected needs to be aggregated.

**Mediating Variable: Boundary-spanning searches.** Drawing on the research of Katila and Ahuja (2002) and Laursen and Salter (2006) and referring to the study of Henttonen and Ritala (2013) on multi-source search and focused search, this study used 4 items to measure acute boundary-spanning search (AS): “We are able to develop a variety of partners and build relationships”, “We are able to identify new opportunities and target new customers in a timely manner”, “We are able to track changes in the market environment in a timely manner”, “We pay close attention to the movements of our competitors and respond quickly”; four items were used to measure concentrated boundary-spanning search (CS): “We are able to maintain close contact with existing partners”, “A few cooperation channels provide us with most of our knowledge”, “We pay attention to the changing needs of the current market”, “We are able to acquire knowledge that can help improve existing product development processes and skills”.

**Dependent variable: Business model innovation.** Drawing on the scale of Zott and Amit (2007) on business model design and the research of Yu et al. (2020), four items were used to measure the novelty-centered business model innovation (BMN) and the efficiency-centered business model innovation (BME). BMN was measured by: our business models: “Provides a new combination of products, services, and information”, “Creates a new way to make money”, “Introduces a new operational process or specification”, “Connects participants to transactions in a novel way”. BME was measured by: our business models: “Reduce information asymmetry in the transaction process”, “Simplify the transaction process”, “Reduce transaction costs”, and “Improve transaction speed”.

**Moderating variable: Environmental dynamism (ED).** According to Jansen et al. (2006), environmental dynamism was measured with 4 items, including: “Our market environment changes dramatically”, “Our customers often request new products or services”, “In our market, changes are constantly happening”, and “The quantity of products and services to be delivered changes rapidly and frequently”.

**Control variables.** Referring to the research of Narayan et al. (2021) at firm level, this study selects the field, firm age, and firm size as control variables. The field of enterprises and their ownership affect firm inertia and thus the performance of business model innovation. At the same time, firm age also affects the performance of business model innovation, as relatively small companies are more likely to be significantly affected by the decisions of top managers, while large companies are more likely to be affected by their ecosystem (Heavey and Simsek, 2015; Lubatkin et al., 2006). In addition, according to the research of Yang et al. (2021) the type of ownership will affect the innovation orientation of enterprises, so this study introduces enterprise ownership as a control variable. At the TMT level, the selected team size was used as a control variable. Larger TMT may bring richer resources to business model innovation. The firm age was coded from 1 to 4 according to the years of establishment (less than 3 years, 3–5 years, 5–10 years, and more than

10 years), and the firm scale was coded from 1 to 4 according to the number of employees (less than 50, 50–200, 200–500, and more than 500); firm ownership was divided into four types, and a dummy variable was set with foreign-owner as the reference. The size of the TMT was measured by the number of members.

### 3.3. Aggregation of variables

The measurement for the transactive memory system was accomplished at the individual level and needs to be aggregated to team level. In accordance with comparable studies (Heavey and Simsek, 2015), within-group interrater reliability (Rwg) (George, 1990; LeBreton and Senter, 2008) and intra-class correlation (ICC) (Chen and Bliese, 2002) were used to evaluate the consistency of team members' answers. For all teams, the Rwg values meet the requirements (RwgMin = 0.72, RwgMax = 0.99, RwgAve = 0.92, RwgMid = 0.94), indicating high consistency among team members. Meanwhile, ICC (1) > 0.15, ICC (2) > 0.5. The results of the two indicators show that individual data can be aggregated to the team level (Chen et al., 2005).

### 3.4. Common method bias

To mitigate the possible effects of common method bias (CMV), the study introduced pre-procedures and post-statistical methods to control for common method bias. In the pre-procedure, the questionnaire for the study consisted of two parts, each part was filled out by different respondents, and the answers of the respondents were guaranteed to remain anonymous and confidential to other respondents. Meanwhile, part A pertained to the transactive memory system, for which the questionnaires were filled out by at least two respondents, and the questionnaire data were aggregated. Then, the order of the items filled in by different respondents was different, so as to reduce the possible impact of the order of the questions. In the post-statistical method, the Harman's signal factor test was used to analyze the influence of common method bias. Specifically, we constrained all of the key variables to one latent construct (Podsakoff et al., 2003), and the test showed that this factor accounted for 29.3% of the total variance, which was below the 50% threshold. Therefore, common method bias is unlikely to be a problem.

### 3.5. Reliability and validity

We used SPSS AMOS 26 software to measure reliability and validity. According to Hair et al. (2021), the factor loadings, average variance extracted (AVE), and combined reliability (CR) of the variables are shown in Table 1. The factor loadings of TMS after aggregation to the team level were all greater than 0.7 (N = 87), and the AVE value was 0.653. The factor loadings of other variables are all greater than 0.7, indicating that the indicators have good reliability (Hair et al., 2012). The Cronbach's  $\alpha$  values of all variables are greater than 0.7 (Clausz, 2017; Nunnally, 1978), and the CR values are greater than 0.6 (Bagozzi and Yi, 1988), indicating that the measurement of variables is reliable. The AVE values for all variables were greater than 0.5 (Fornell and Larcker, 1981), indicating that the measure had good convergent validity. At the same time, according to Table 2, the correlation coefficient between each variable is less than the square root of their AVE (Pedersen et al., 2018), indicating that the variables have good discriminant validity. In confirmatory factor analysis, all items were loaded onto their corresponding latent structures and the 6-factor model fit was good ( $\chi^2$  (362) = 1.342,  $p < 0.001$ , CFI = 0.908, RMSEA = 0.063). At the same time, the 6-factor model fits better than other competing models. Therefore, it can be considered that each variable in the model has sufficient validity and can be distinguished from each other.

**Table 1**  
Reliability and construct validity.

| Variables    | Items                                                                                                       | Factor loading | AVE  | $\alpha$ | CR   |
|--------------|-------------------------------------------------------------------------------------------------------------|----------------|------|----------|------|
| TMS (N = 87) | Each team member has specialized knowledge of some aspect of our work                                       | 0.78           | 0.65 | 0.93     | 0.94 |
|              | The specialized knowledge of several team members is needed to achieve our goals                            | 0.78           |      |          |      |
|              | I know which team members have expertise in specific areas                                                  | 0.81           |      |          |      |
|              | I am comfortable accepting suggestions from other team members.                                             | 0.82           |      |          |      |
|              | I trust that other members' knowledge is credible.                                                          | 0.80           |      |          |      |
|              | I am confident relying on the information that other team members bring to the discussion                   | 0.82           |      |          |      |
|              | Our team works together in a well-coordinated fashion                                                       | 0.79           |      |          |      |
|              | Our team has very few misunderstandings about what to do                                                    | 0.83           |      |          |      |
|              | We tend to accomplish our work smoothly and efficiently                                                     | 0.83           |      |          |      |
|              | We can develop a variety of partners and establish relationships                                            | 0.75           |      |          |      |
| AS (N = 87)  | we can identify new opportunities and locate new customers in time                                          | 0.80           | 0.62 | 0.80     | 0.87 |
|              | we can track changes in the market environment in time                                                      | 0.82           |      |          |      |
|              | we pay close attention to the trends of our competitors and respond quickly                                 | 0.78           |      |          |      |
|              | We can keep close contact with existing partners                                                            | 0.80           |      |          |      |
| CS (N = 87)  | a few cooperation channels provide us with most of the knowledge                                            | 0.79           | 0.61 | 0.79     | 0.86 |
|              | we pay attention to the demand changes in the current market                                                | 0.77           |      |          |      |
|              | we can acquire knowledge that is conducive to improving the existing product development process and skills | 0.77           |      |          |      |
|              | Environmental changes in our local market are intense                                                       | 0.82           |      |          |      |
| ED (N = 87)  | Our clients regularly ask for new products and services                                                     | 0.84           | 0.70 | 0.86     | 0.90 |
|              | In our local market, changes are taking place continuously                                                  | 0.86           |      |          |      |
|              | In our market, the volumes of products and services to be delivered change fast and often                   | 0.83           |      |          |      |
|              | the business model offers new combinations of products, services, and information                           | 0.82           |      |          |      |
| BMN (N = 87) | the business model creates a new profitable way                                                             | 0.82           | 0.70 | 0.80     | 0.90 |
|              | the business models introduces new operational processes or specifications                                  | 0.85           |      |          |      |
|              | the business model links participants to transactions in novel ways                                         | 0.84           |      |          |      |
|              | the business model reduces information asymmetry in the transaction process                                 | 0.81           |      |          |      |
| BME (N = 87) | the business model simplifies the transaction process                                                       | 0.78           | 0.62 | 0.86     | 0.87 |
|              | the business model reduces transaction costs                                                                | 0.72           |      |          |      |
|              | the business model supports fast transactions                                                               | 0.84           |      |          |      |

## 4. Results

The mean, standard deviation, and variable correlation results for the variables are shown in Table 2. The TMS in Table 2 was derived from questionnaire part A and aggregated to the TMT level; other variables, including control variables, were derived from questionnaire B. There is a certain correlation between the variables, which provides a precondition for the subsequent analysis.

### 4.1. Main effects

We tested the hypothesis using hierarchical regression analysis and the data were normalized using means. According to the regression results shown in Table 3, the relationship between TMS and BMN was positive and significant (model 6,  $\beta = 0.244$ ,  $p < 0.05$ ) and the relationship between TMS and BME was also positive and significant (model 11,  $\beta = 0.268$ ,  $p < 0.05$ ), indicating that the direct relationships were significant. Thus, H1a and H1b were supported.

### 4.2. Indirect effects

According to the BK method (Baron and Kenny, 1986), hierarchical regressions were used to test the mediation effects. As shown in Table 3, there was a positive and significant relationship between TMS and AS (model 2,  $\beta = 0.386$ ,  $p < 0.01$ ) and a positive and significant relationship between TMS and CS (model 4,  $\beta = 0.389$ ,  $p < 0.01$ ), indicating that the relationships between the independent variable and the mediating variables were significant. Both H2a and H2b were supported. There was a significant and positive relationship between AS and BMN (model 7,  $\beta = 0.409$ ,  $p < 0.01$ ) and a significant and positive relationship between CS and BME (model 12,  $\beta = 0.523$ ,  $p < 0.01$ ). Obviously, H3a and H3b were supported. Therefore, it can be preliminarily considered that AS plays a mediating role between TMS and BMN, and CS plays a mediating role between TMS and BME, and both H4a and H4b were supported.

### 4.3. Moderating effects

For BMN, after adding the moderating variable ED and its interaction with AS, the regression coefficient of the interaction was positive and significant (model 9,  $\beta = 0.180$ ,  $p < 0.05$ ) and the coefficient on the mediating variable AS was significantly positive (model 9,  $\beta = 0.434$ ,  $p < 0.01$ ). Thus, ED significantly enhanced the relationship between AS and BMN, hypothesis H5a was supported. For BME, after adding the moderating variable ED and its interaction with CS, the regression coefficient of the interaction was negative and significant (model 14,  $\beta = -0.224$ ,  $p < 0.05$ ), and the coefficient on the mediating variable CS was positive and significant (model 14,  $\beta = 0.520$ ,  $p < 0.01$ ). Thus, ED significantly weakened the relationship between CS and BME, hypothesis H5b was supported.

Bootstrap analyses were performed using PROCESS for SPSS software to test the direct effects and conditional indirect effects of the model. A total of 1000 bootstrap samples were performed, and the 95% bias-corrected confidence intervals are shown in Table 4.

For the relationship between TMS and BMN, the 95% confidence interval for the indirect effect did not contain 0 ( $B = 0.239$ ,  $CI = [0.071, 0.466]$ ), further indicating that AS played a mediating role between TMS and BMN. For the relationship between TMS and BME, the 95% confidence interval for the indirect effect did not contain 0 ( $B = 0.270$ ,  $CI = [0.090, 0.513]$ ), further indicating that CS played a mediating role between TMS and BME. Therefore, both hypotheses H4a and H4b were further supported.

As for the moderated mediating effect of TMS on BMN via AS moderated by ED, the 95% confidence interval for the conditional indirect effect contained 0 ( $B = 0.158$ ,  $CI = [-0.002, 0.401]$ ) when ED was at a lower value (Mean-1SD); and the 95% confidence interval for the conditional indirect effect when ED was at a higher value (Mean and

**Table 2**

Means, standard deviations, and correlations.

| variables  | 1        | 2        | 3     | 4        | 5       | 6       | 7     | 8       | 9           | 10          | 11          | 12          | 13          |
|------------|----------|----------|-------|----------|---------|---------|-------|---------|-------------|-------------|-------------|-------------|-------------|
| ownership1 |          |          |       |          |         |         |       |         |             |             |             |             |             |
| ownership2 | −0.55*** |          |       |          |         |         |       |         |             |             |             |             |             |
| ownership3 | −0.44*** | −0.32*** |       |          |         |         |       |         |             |             |             |             |             |
| industry   | −0.18*   | −0.05    | 0.18* |          |         |         |       |         |             |             |             |             |             |
| firm age   | 0.00     | −0.12    | −0.14 | −0.36*** |         |         |       |         |             |             |             |             |             |
| firm size  | −0.16    | 0.23**   | −0.10 | −0.23**  | 0.45*** |         |       |         |             |             |             |             |             |
| TMT size   | −0.07    | 0.16     | −0.14 | 0.05     | 0.15    | 0.49*** |       |         |             |             |             |             |             |
| TMS        | 0.08     | 0.06     | −0.01 | 0.09     | −0.07   | −0.05   | −0.08 | 0.67    |             |             |             |             |             |
| AS         | 0.00     | 0.19*    | −0.13 | 0.27**   | −0.25** | −0.01   | 0.03  | 0.42*** | <b>0.79</b> |             |             |             |             |
| CS         | 0.02     | 0.12     | 0.06  | 0.13     | −0.23** | −0.09   | 0.06  | 0.44*** | 0.36***     | <b>0.78</b> |             |             |             |
| BMN        | −0.04    | 0.13     | −0.06 | 0.25**   | −0.07   | 0.02    | −0.04 | 0.29*** | 0.49***     | 0.27**      | <b>0.84</b> |             |             |
| BME        | −0.10    | 0.17     | −0.01 | 0.06     | −0.22** | −0.01   | 0.10  | 0.24**  | 0.41***     | 0.53***     | 0.31***     | <b>0.83</b> |             |
| ED         | 0.15     | −0.18*   | −0.07 | 0.16     | 0.20*   | 0.12    | −0.07 | −0.04   | −0.06       | −0.27**     | 0.53***     | −0.30***    | <b>0.79</b> |
| Mean       | 0.43     | 0.29     | 0.21  | 2.61     | 2.22    | 2.64    | 5.59  | 3.40    | 3.25        | 3.20        | 3.33        | 3.37        | 3.50        |
| S.D.       | 0.50     | 0.46     | 0.41  | 1.09     | 0.99    | 0.99    | 1.99  | 0.71    | 1.01        | 0.93        | 1.07        | 0.94        | 1.11        |

NOTE: \* $P < 0.1$ , \*\* $P < 0.05$ , \*\*\* $P < 0.01$ ; Numbers in bold show the square-root of the AVE.

Mean+1SD) did not contain 0. This suggests that the mediating relationship would not be affected at lower ED values. It also indicates that the indirect effect increases with the increase of ED. Therefore, hypothesis H6a was partially supported. As for the moderated mediating effect of TMS on BME via CS moderated by ED, the 95% confidence interval of the conditional indirect effect did not contain 0 when the ED was at a lower value (Mean -1SD) to the Mean, and the 95% confidence interval of the conditional indirect effect contained 0 ( $B = 0.1478$ ,  $CI = [-0.0254, 0.3991]$ ) when ED was at a higher value (Mean + 1SD). This suggests that the mediating relationship would be affected when the ED is at a medium or low value, and that the mediating relationship is not significant when the ED is at a higher value. The indirect effect values decrease as the ED increases. Therefore, hypothesis H6b was partially validated.

## 5. Discussion

Based on organizational learning theory, this study explored the impact of the transactive memory system of the top management teams on the dual attributes of enterprise business model innovation, as well as the mediating effect of two boundary-spanning searches and the moderating effect of environmental dynamism. Specifically, we divided boundary-spanning searches into acute (AS) and concentrated search (CS) according to the attention allocation of search behavior. Meanwhile, the impacts of the two boundary-spanning searches on novelty-centered and efficiency-centered business model innovation were introduced. Furthermore, the study explored the moderating effect of environmental dynamism on the mediating relationships. Our hypothesis was tested by analyzing the questionnaire data gathered from 87 Chinese gazelle enterprises in different fields. It was found that transactive memory system had significant and positive impacts on acute boundary-spanning search, concentrated boundary-spanning search, novelty-centered business model innovation, and efficiency-centered business model innovation; acute boundary-spanning search and concentrated boundary-spanning search played positive mediating roles in the relationship between transactive memory system and novelty-centered and efficiency-centered business model innovation, respectively. Meanwhile, environmental dynamism played a differential moderating role in the relationship between boundary-spanning searches and business model innovation; it positively moderated the relationship between acute boundary-spanning search and novelty-centered business model innovation but negatively moderated the relationship between concentrated boundary-spanning search and efficiency-centered business model innovation. Environmental dynamism also had a moderating effect on the mediating relationships. The stronger the environmental dynamism, the stronger the relationship between transactive memory system influencing novelty-centered

business model innovation via acute boundary-spanning search and the weaker the relationship between efficiency-centered business model innovation via concentrated boundary-spanning search. Our study has important theoretical and managerial significance.

### 5.1. Theoretical implications

Empirical research on the internal drivers of business model innovation has attracted increasing attention, and the impacts of top managers have become an important research topic in business model innovation due to their decisive role in organizational decision-making. Existing literature has explored the drivers of business model innovation in terms of TMT's behavior, experience, and team background. However, few studies have considered the mechanisms by which the knowledge perception factors of the TMT influence business model innovation. This study makes contributions to the literature in the following three ways.

First, our study enhances the antecedent literature on business model innovation. There is a strong call for increased research on the antecedents of business model innovation, especially internal antecedents (Foss and Saebi, 2017; Snihur and Wiklund, 2019). Our efforts on the relationship between transactive memory system and business model innovation are a useful exploration of research on the antecedents of business model innovation. More importantly, by validating the mediating role of boundary-spanning searches between transactive memory system and business model innovations, this study complements the mechanism of the influence of internal and external learning on business model innovation performance in organizational learning by providing a way for internal and external learning to act together from the behavior of the top management team. While recent research has explored the direct effects of interactive memory systems on different attributes of business models (Chen and Yi, 2023), the underlying mediating mechanisms of the relationship between interactive memory systems and business model innovation have not been fully explored. Our study provides a valuable exploration of the black box between the relationship between interactive memory systems and business model innovation. At the same time, the mediating role of boundary-spanning search revealed by the study provides new perspectives for understanding the relationship between internal and external learning. At the same time, the environment is an external factor that is closely related to the business model (Peprah et al., 2022; Yuan et al., 2021), our study explains the important moderating role of environmental dynamics in mediating relationships. The joint effects of boundary-spanning searches and environmental factors on business model innovation have been revealed. Thus, our study provides a more inclusive structure that can inform subsequent research on the antecedents of business model innovation. In conclusion, this study echoes the research calls of

**Table 3**  
Results for testing hypotheses.

| variable                | AS      |          |          | CS       |          |         | BMN      |           |           | BME      |          |          |
|-------------------------|---------|----------|----------|----------|----------|---------|----------|-----------|-----------|----------|----------|----------|
|                         | model 1 | model 2  | model 3  | model 4  | model 5  | model 6 | model 7  | model 8   | model 9   | model 10 | model 11 | model 12 |
| ownership1              | 0.203   | -0.022   | 0.583**  | 0.356    | 0.252    | 0.109   | 0.119    | 0.002     | -0.065    | -0.097   | -0.254   | -0.440** |
| ownership2              | 0.265   | 0.050    | 0.610*** | 0.394*   | 0.324    | 0.188   | 0.168    | 0.204     | 0.140     | 0.050    | -0.099   | -0.305   |
| ownership3              | -0.022  | -0.176   | 0.487**  | 0.332*   | 0.094    | -0.003  | 0.069    | 0.104     | 0.076     | -0.049   | -0.155   | -0.329*  |
| industry                | 0.301** | 0.223*   | 0.151    | 0.073    | 0.361*** | 0.312** | 0.220*   | -0.007    | -0.031    | -0.050   | -0.103   | -0.141   |
| firm age                | -0.161  | -0.219*  | -0.007   | -0.065   | 0.089    | 0.052   | 0.142    | 0.003     | -0.029    | -0.260*  | -0.300** | -0.266** |
| firm size               | 0.125   | 0.128    | -0.107   | -0.103   | 0.102    | 0.104   | 0.052    | -0.100    | -0.104    | -0.001   | 0.001    | 0.055    |
| TMT size                | -0.050  | -0.012   | 0.117    | 0.155    | -0.137   | -0.113  | -0.108   | 0.030     | 0.049     | 0.124    | 0.150    | 0.069    |
| TMS                     |         | 0.386*** |          | 0.389*** |          | 0.244** | 0.409*** | 0.452***  | 0.434***  |          | 0.268**  | 0.064    |
| AS                      |         |          |          |          |          |         |          |           |           |          |          | 0.523*** |
| CS                      |         |          |          |          |          |         |          |           |           |          |          | 0.490*** |
| ED                      |         |          |          |          |          |         |          |           |           |          |          | -0.128   |
| ED*AS                   |         |          |          |          |          |         |          |           |           |          |          | -0.224** |
| ED*CS                   |         |          |          |          |          |         |          |           |           |          |          | 0.360    |
| R <sup>2</sup>          | 0.171   | 0.306    | 0.150    | 0.288    | 0.118    | 0.172   | 0.288    | 0.594     | 0.622     | 0.087    | 0.152    | 0.348    |
| adjusted R <sup>2</sup> | 0.097   | 0.235    | 0.075    | 0.214    | 0.040    | 0.087   | 0.205    | 0.540     | 0.566     | 0.006    | 0.065    | 0.271    |
| F                       | 2.322** | 4.299*** | 1.994*   | 3.934*** | 1.512    | 2.027*  | 3.467*** | 11.100*** | 11.205*** | 1.078    | 1.753*   | 4.560*** |

NOTE: \* $P < 0.1$ , \*\* $P < 0.05$ , \*\*\* $P < 0.01$ .**Table 4**  
Results for bootstrapping.

|    | variable                     | effect | BootSE | BootLLCI | BootULCI |
|----|------------------------------|--------|--------|----------|----------|
| AS | Indirect effect(s)           | 0.239  | 0.099  | 0.071    | 0.466    |
|    | ED                           | 0.158  | 0.103  | -0.002   | 0.401    |
|    | Conditional indirect effects | 0.254  | 0.093  | 0.089    | 0.463    |
| CS | Indirect effect(s)           | 0.349  | 0.107  | 0.152    | 0.573    |
|    | ED                           | 0.270  | 0.107  | 0.090    | 0.513    |
|    | Conditional indirect effects | 0.387  | 0.174  | 0.117    | 0.798    |
|    |                              | 0.267  | 0.116  | 0.085    | 0.534    |
|    |                              | 0.148  | 0.111  | -0.025   | 0.399    |

Note: condition values: Mean + 1SD, Mean, and Mean - 1SD.

Chesbrough (2010) and Foss and Saebi (2017) on the antecedents of business model innovation.

Second, we have refined the organizational learning theory by introducing the concept of attention allocation to boundary-spanning search. Based on organizational learning theory, we explored the impact of the acquisition behavior of external heterogeneous knowledge resources across organizational boundaries on business model innovation. Existed studies have explored the impact of the search content, location and timing attributes of boundary-spanning search on the performance of firm innovation (Katila and Ahuja, 2002; Lopez-Vega et al., 2016; Snihur and Wiklund, 2019; Wang et al., 2020). However, when considering the characteristics of the search subject, the attentional characteristics of the perpetrators of boundary-spanning search behavior have been largely neglected. Attention is an important resource for organizations (Ocasio, 2011), and differences in attention allocation may be a source of differences in firms' innovation performance (Henttonen and Ritala, 2013). In this study, by considering the attention allocation of organizational search behavior, boundary-spanning search was divided into acute boundary-spanning search and concentrated boundary-spanning search, and models of their impacts on the dual attributes of business model innovation were constructed. This provides a new pathway for future research. It is also a useful addition to the research framework by Berends et al. (2016) and Yuan et al. (2021) on the relationship between learning duality and the dual attributes of business model innovation. The joint effects of boundary-spanning searches and environmental factors on business model innovation have been revealed. We found that the impact of boundary-spanning searches on business model innovations will vary with the attention characteristics of boundary-spanning searches, business model innovation attributes and ED. This finding also provides further evidence for the idea that organizational learning is closely related to the external environment (Argote et al., 2021).

Third, we contribute to the transactive memory system literature and a new explanation for the impact of TMT's diversity on business model innovation is given. The significance of top managers and their teams in a firm's innovation behaviors, such as product innovation, process innovation, and knowledge innovation, has been widely argued (Boone et al., 2019; Garms and Engelen, 2019; Narayan et al., 2021; Talke et al., 2010). At the same time, the diversity of knowledge in TMT is an important driver of innovation (Qian et al., 2013). On this basis, the influence of team cognitive factors on business model innovation has begun to receive attention (Snihur and Zott, 2020). Transactive memory system is an important team cognitive characteristic (Mohammed et al., 2021). However, it has been largely understudied in relation to this key organizational outcome, which is business model innovation. Existing research has demonstrated the effects of transactive memory system on team creativity, product innovation, and team or organizational performance (Bachrach et al., 2019; Bachrach and Mullins, 2019; Mohammed et al., 2021; Sáiz-Pardo et al., 2021; Tang et al., 2015). Taking the above research a step further, this study revealed the significance of transactive memory system as an important antecedent to

boundary-spanning searches, which has been ignored in previous studies. In a context where the pursuit of diversity but its extent and threshold are highly debated, our study draws on the concept of transactive memory system to give a meaningful explanation that transactive memory system contributes to organizational learning and positively influences firms' innovative behavior. Our theoretical model of the role of the TMT's transactive memory system on the novelty and efficiency attributes of business model innovation and empirically validated the positive impact of transactive memory system on the two attributes of business model innovation. This is also in line with research on the positive effects of transactive memory system on creativity and innovation performance (Bachrach et al., 2019; Wegner, 1987). At the same time, this study has suggested that transactive memory system has a positive impact on the organization's boundary-spanning search behavior, which also validates the positive impact of transactive memory system on the organization's collective learning process (Akgun et al., 2006). This research also validates the view of Heavey and Simsek (2015) that the beneficial effects of transactive memory system are inseparable from external sources of knowledge. Lastly, most previous studies on transactive memory system have mostly involved setting up experimental conditions and obtaining data manually, or forming ad hoc teams for research purposes to obtain data. Our results generated by empirical evidence confirm the applicability of transactive memory system in practical management teams, especially top management teams. This not only extends the application of transactive memory system in team research, but also extends the study of TMT's cognition.

### 5.2. Practical implications

Our research yields some managerial implications. First, transactive memory system is an important resource for companies, and those seeking to innovate their business models should focus on building and developing transactive memory system for their teams. A TMT committed to business model innovation can certainly pursue the diversity of team characteristics, and the role of such diversity for innovation has been emphasized. However, managers must now recognize that the transactive memory system, a diversity-based team cognition, is a direct driver of business model innovation. TMTs should pursue a more refined transactive memory system to improve learning and thus facilitate business model innovation. For example, optimizing the knowledge structure of team members and building a cross-disciplinary talent knowledge base. Specifically, specialization, trust, and coordination are all important responsive forms of transactive memory system. It should be noted that when forming or replenishing TMT's, Enterprises should actively improve the division of labor in TMT through the specialization of TMT members. On the one hand, relevant incentive policies need to be developed to strengthen trust and communication among members and integrate knowledge and expertise among members. On the other hand, managers can use the interactive memory system to diagnose the degree of integration of members' knowledge and expertise to help inefficient teams improve their innovation effectiveness.

Our research provides new possible practice paths for senior managerial teams to innovate their business models through learning behaviors. The study found that while both attributes of business model innovation benefit from boundary-spanning search, there are different paths of influence. Therefore, TMTs should first determine the characteristics of their own business model and identify the direction or preference of business model innovation. Managers have to carry out relevant boundary-spanning search behaviors according to the attributes of business model innovation. Meanwhile, our research emphasizes importance of the mediating role of boundary-spanning searches. Enterprises' TMT can compensate for the limitations of their internal cognitive framework through external learning, and new knowledge and skills can be learned through boundary-spanning search to compensate for the limitations of members' cognition of existing resources. Pursuing a novelty-centered business model innovation needs to focus more on

achieving acute search, that is, acquiring knowledge in a timely manner through a wider range of knowledge sources; an efficiency-centered business model innovation should focus on concentrated search, that is, efficiently searching for knowledge through specific knowledge source channels. Our research underscores the importance of senior managers accurately judging changes in the external environment as a contextual condition. In addition, managers should be clear that the specific degree to which firms conduct acute boundary-spanning search or focus on boundary-spanning search should be spontaneously adjusted as the environment changes. In an environment with frequent changes, TMTs pursuing novel business model innovation should focus on broader knowledge sources and the timeliness of information acquisition. And in an environment with relatively stable environment, TMTs pursuing efficient business model innovation focus on existing knowledge source channels to acquire and utilize knowledge more effectively.

In general, our research provides more possible paths for companies and top management teams to innovate their business models. As business model innovation has become a more important source of value in some booming economies (Luo et al., 2022), this will not only help the focal company create more value, but also provide more equal and increased value for every participant in the business models.

### 5.3. Limitations and future research

This study has the following limitations: Firstly, we have several limitations with regard to the sample data. The first limitation exists in terms of the procedures used to obtain the sample data. For the sample data acquisition procedures, we conducted some procedures in place to control for CMV, but our data was derived from self-reports by respondents, and both parts of the questionnaire were completed at the same time, so there might be other biases. Also, the sample size of the study was small. Although it met the minimum number requirement and did not cause obvious distress to the analyses, future studies should expand the sample size to further examine whether our model works. A second limitation resides in the representativeness of the sample. Our sample data was gathered from Chinese companies. However, whether the research conclusions can be extended to other countries or regions remains to be seen. The applicability of the findings can be verified in the future by studying samples from different institutions and cultures in other regions. Furthermore, cross-sectional data cannot reflect causal relationships between variables. We suggest that future studies can be conducted by following data at multiple points in time, or by using objective panel data to verify possible dynamic relationships, or by using case studies to further validate the robustness of our analysis.

Our research also implies the following possible research directions: first, there may be other competing mediating relationships. For example, resource orchestration capacity is an important factor affecting the differences in outcomes between firms (Sirmon et al., 2007, 2011). It explains the mechanisms by which resource acquisition from the external environment and subsequent internal allocation are related to firm outcomes (Abdurakhmonov et al., 2021; Adner and Helfat, 2003). Second, the relationships between transactive memory system, boundary-spanning searches, and business model innovations are also affected by multiple team and organizational factors. It is possible that there are other moderating factors. Therefore, it is possible to explore the moderating effects of factors such as the cultural background of TMT, the innovation atmosphere at the team or enterprise level, and the strategic orientation or values of the enterprise. Finally, we think that the drivers of business model innovation should be further considered using a network-based perspective. The connectivity properties of business models suggest that focal enterprises are part of their cooperative networks, which can provide relevant channels for knowledge acquisition and promote knowledge flow (Dyer and Nobeoka, 2000; Hock-Doepgen et al., 2021). For example, the influence of network location on problem search has been demonstrated (Hu et al., 2021). Therefore, considering the network of enterprises may bring new

insights to business model innovation.

## Data availability

Data will be made available on request.

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